Paleomagnetism of the Borden dykes of Baffin Island and its bearing on the Grenville Loop

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Northern Baffin Island is intersected by spectacular swarms of predominantly northwesterly trending diabase dykes. Cross-cutting relationships, K–Ar ages, and paleomagnetic contact tests indicate that these dykes represent at least two major episodes of Hadrynian igneous activity. The Borden dykes were emplaced during the earlier of these episodes. They are about 950 Ma old and have a paleomagnetic pole position of 153.3°E, 26.7°S, $\delta m = 8.6°$, $\delta p = 6.1°$. The later episode saw the emplacement of the previously defined Franklin dykes, which are about 750 Ma old. The newly defined Borden pole lies at the southern extremity of the Grenville Loop and has an age consistent with poles of similar position that were derived from rocks of the Grenville Province.

All of the dykes probably reflect tension acting in an east-northeast–west-southwest direction. The directions of thick northwest trending dykes, which form a considerable angle to that expected from east-northeast tension, are thought to have been controlled by earlier (circa 1200 Ma) fractures of the Borden fault zones. The two main surges of magma may coincide with two periods of Hadrynian uplift in the Boothia Peninsula area to the west.

Dans le nord de l'île de Baffin il existe un réseau très apparent de dykes de diabase qui s'entrecoupent et dont l'orientation prédominante présente une direction nord-ouest. Les entrecoupements relatifs, les âges mesurés au K–Ar et les déterminations paléomagnétiques dans les zones de contact révèlent que ces dykes représentent au moins deux épisodes majeurs d'activité ignée à l'Hadrynien. La mise en place des dykes Borden s'est produite dès le début de ces épisodes. Leur âge est approximativement 950 Ma et ils possèdent un pôle paléomagnétique 153,3°E, 26,7°S, $\delta m = 8,6°$, $\delta p = 6,1°$. Le dernier épisode fut témoin de la mise en place des dykes Franklin décrits antérieurement et datés à environ 750 Ma. Le pôle Borden nouvellement défini se situe à l'extrémité sud de la boucle du Grenville et son âge concorde avec les pôles occupant une position semblable déterminés à partir des roches de la province de Grenville.

L'ensemble des dykes reflète possiblement un système de tension ayant agi dans une direction est-nord-est–ouest-sud-ouest. Les directions des dykes épais orientés nord-ouest formant un angle prononcé avec les directions des tensions, vraisemblablement orientées est-nord-est, semblent avoir été influencées par des fractures antérieures (environ 1200 Ma) associées aux zones de la faille Borden. Les deux principales montées de magma peuvent coincider avec deux périodes de soulèvement à l'Hadrynien dans la région de la péninsule de Boothia à l'ouest.

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Introduction

Early paleomagnetic work (Fahrig *et al.* 1971; Fahrig and Schwarz 1973) on the great swarms of diabase dykes that extend along the axis of Baffin Island resulted in the identification of a large number of dykes that exhibit an apparently primary, stable, gently dipping southeasterly (or reversed) magnetization. These dykes are about 750 Ma old and were correlated with basic intrusions of similar character elsewhere in the Canadian Shield. The intrusions were referred to collectively as Franklin intrusions.

Close to, and apparently parallel with the Baffin Island dykes exhibiting Franklin characteristics, were recognized a number of dykes with a steep, southeast magnetization. These were thought to be remagnetized Franklin dykes because they are less stably magnetized; their magnetization direction lies in a plane that includes both the Franklin direction and recent Earth's field direction, and several occur towards the northeast coast of Baffin Island where they are more likely to have been heated during the early opening of Baffin Bay. Difficulty in isolating a primary stable component in samples from these dykes was probably exacerbated by inadequacies in the magnetometer used in the 1970 work.

Work has now been carried out on 291 sun-oriented samples from 42 sites on the extension of these swarms on Borden Peninsula in order to clarify the relationship between the two paleomagnetic directions.

Dyke intersections

The basic dykes of Borden Peninsula exhibit two major trends. The more spectacular northwesterly trending swarm is made up of dykes up to 250 m in thickness and 100 km in length. Dykes of the less prominent north-northwesterly trend, with few exceptions, are thinner and shorter. These north-northwesterly dykes are commonly refracted in the main zone of northwesterly dykes (Fig. 1). This would suggest that the

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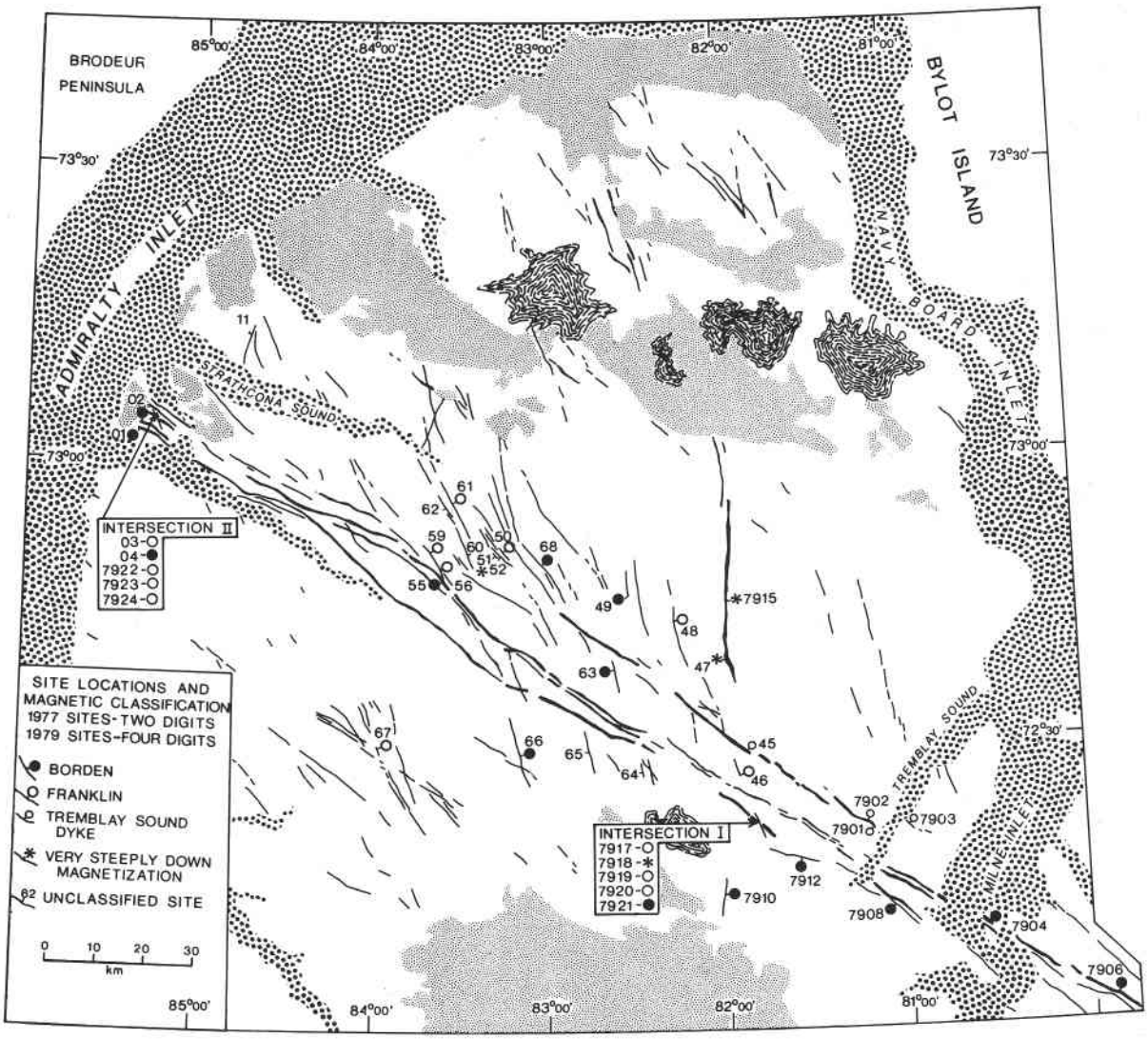


FIG. 1. The distribution of diabase dyke swarms on Borden Peninsula and their paleomagnetic classification. Dyke distribution largely from Blackadar (1968).

north-northwesterly swarm is the younger; however, this zone of refraction also parallels the main Borden fault zones (Jackson *et al.* 1978), which form a pre-existing, deep, fracture system. In any case, north-northwesterly dykes were observed to cut thicker northwest trending dykes at two localities, and rather detailed paleomagnetic sampling was carried out at these localities.

Intersection I

At this intersection (Fig. 2) a north-northwest dyke 25 m thick cuts a northwest dyke that is about 200 m thick. Outcrop is almost lacking in the actual zone of intersection; however, a few metres of the contact between the two dykes is exposed. Seven cores were drilled in the older dyke (site 7919)[1] less than 5 m from the contact, and four were drilled from the chilled marginal zone of the younger dyke (site 7920). Sites 7918 and 7921 were drilled in the older dyke 300 and 65 m, respectively, from the younger dyke. Site 7917 (Figs. 3, 6) was drilled in the younger dyke 250 m from the older. The paleomagnetic results as shown in Fig. 3 provide an excellent contact test indicating that the steep southeasterly paleomagnetic direction is older than the

[1]Throughout this paper, sites from which material was collected in 1979 are prefixed 79. The remaining sites have no prefix and these sites were sampled in 1977.

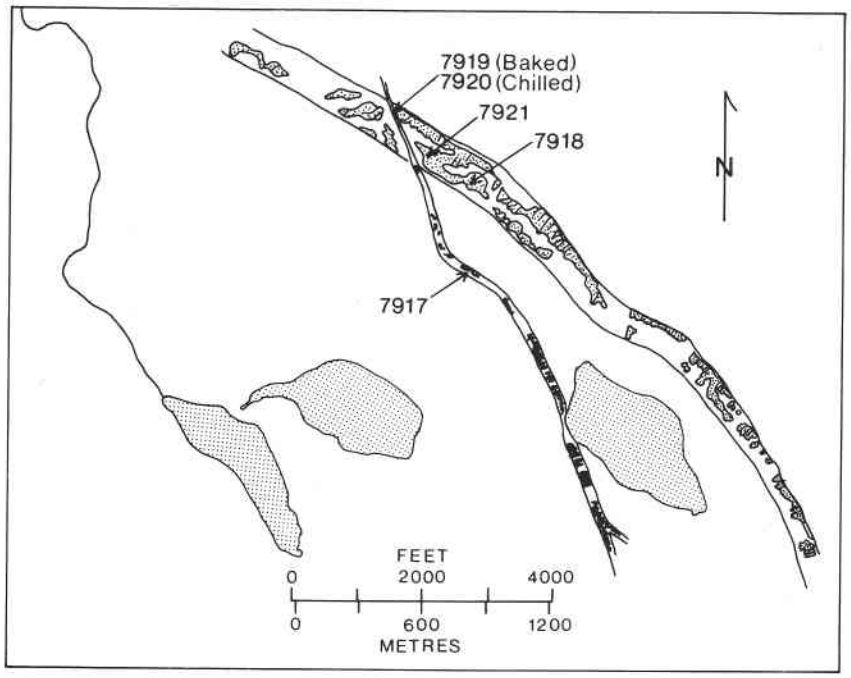


FIG. 2. Detailed sketch from aerial photograph of dyke intersection I (Fig. 1). Outcrops of narrow cross-cutting dykes are shown in solid black, those of the wide northwesterly dyke in stipple.

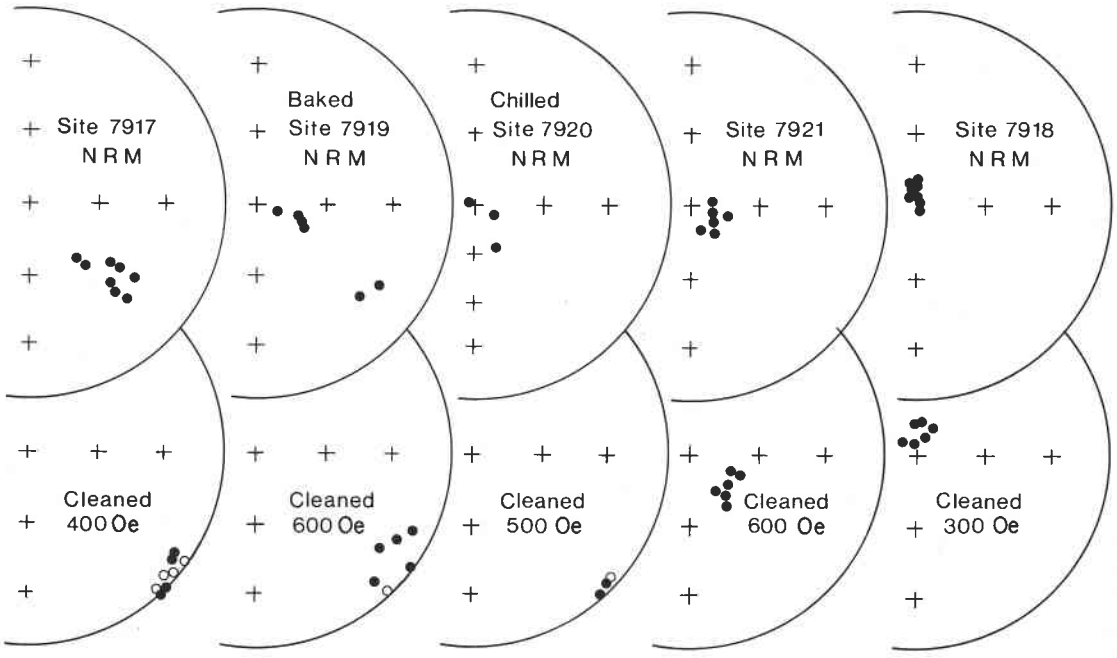


FIG. 3. AF paleomagnetic results from intersection I. In this, and in subsequent stereographic projections, open circles indicate that north-seeking directions are up, solid circles indicate north-seeking directions are down.

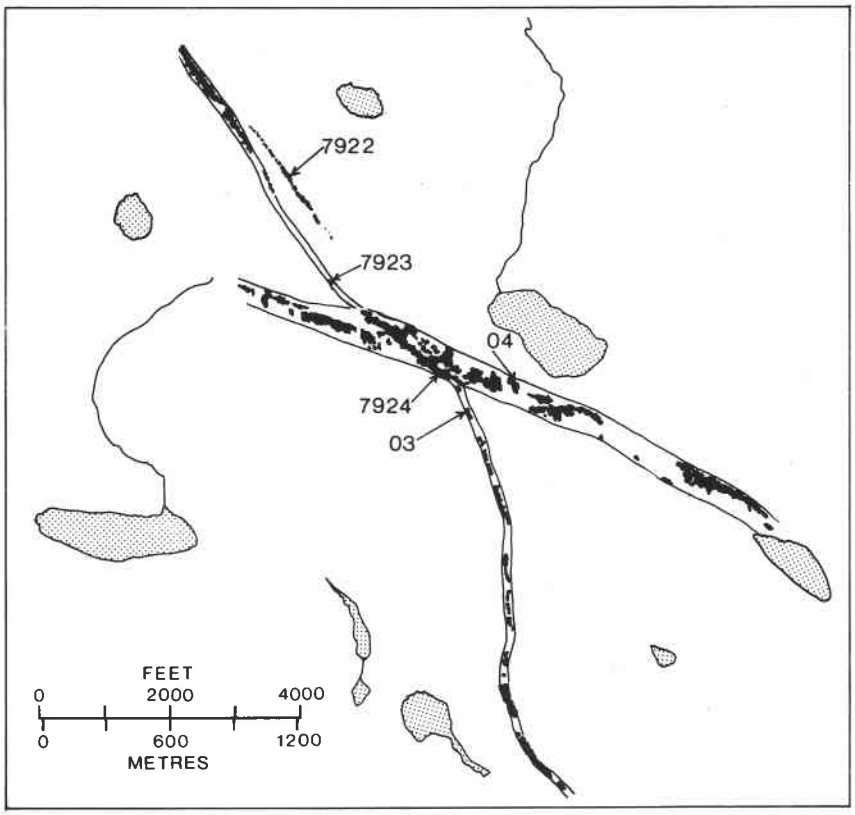


FIG. 4. Detailed sketch, from aerial photograph, of intersection II showing sampling sites. Solid black areas indicate fairly continuous outcrop.

flat southeast Franklin direction and certainly was not, as previously postulated (Fahrig *et al.* 1971), caused entirely by remagnetization of Franklin dykes during the Phanerozoic. Baked site 7919 is of special interest. The distance of samples 1–6 from the younger dyke ranges from 5 m (sample 1) to less than $\frac{1}{3}$ m (samples 5 and 6). The cores farthest from the younger dyke are magnetic hybrids from which the thermoremanent magnetization (TRM) component of the older dyke cannot quite be removed by alternating field (AF) demagnetization (Fig. 3). Chilled marginal material from the younger dyke yielded a whole-rock K–Ar age of 649 ± 42 Ma, K = 0.71. As unaltered, chilled, dyke material generally gives whole-rock K–Ar ages that are 10–15% less than the age of intrusion, this age, along with the paleomagnetic results from this dyke, strongly suggests that this younger north-northwest dyke is a Franklin intrusion. Until Rb–Sr ages are obtained on Franklin intrusions the best estimate for the age of this intrusive event is 750 Ma. Note, in regard to K–Ar dating of basic dykes, that coarser grained material from such dykes generally yields whole-rock K–Ar ages much younger than the age of intrusion (see, for example, dyke ages in Jackson and Iannelli 1981), in some cases up to 30%. There is therefore limited value in obtaining whole-rock K–Ar analyses of coarse diabase to use as a tool in the correlation of individual dykes. On the other hand, biotite extracted from unaltered dykes or from country rock immediately adjacent to wide dykes usually gives ages less than, but often close to, the age of intrusion and similar to ages obtained from chilled material.

The 28° difference between the magnetization directions of sites 7921 and 7918 (Figs. 2, 3) from the wide northwesterly trending dyke is puzzling. The cores of site 7921 were collected at least 60 m away from the younger dyke—too far away to have been affected by this dyke, which has a maximum width of 30 m. Paleosecular variation could account for part of the difference in the two directions. The northwesterly dyke is about 220 m thick and the sampling at each site may represent only 25% of the dyke width. The two sets of samples at the two sites may therefore have passed through their blocking temperatures at times 100 or more years apart. It is unlikely that this explanation would account for all of the observed difference in direction between the two sites. It is more likely that the magnetization displayed by 7918 is not entirely TRM. This is discussed in a later section.

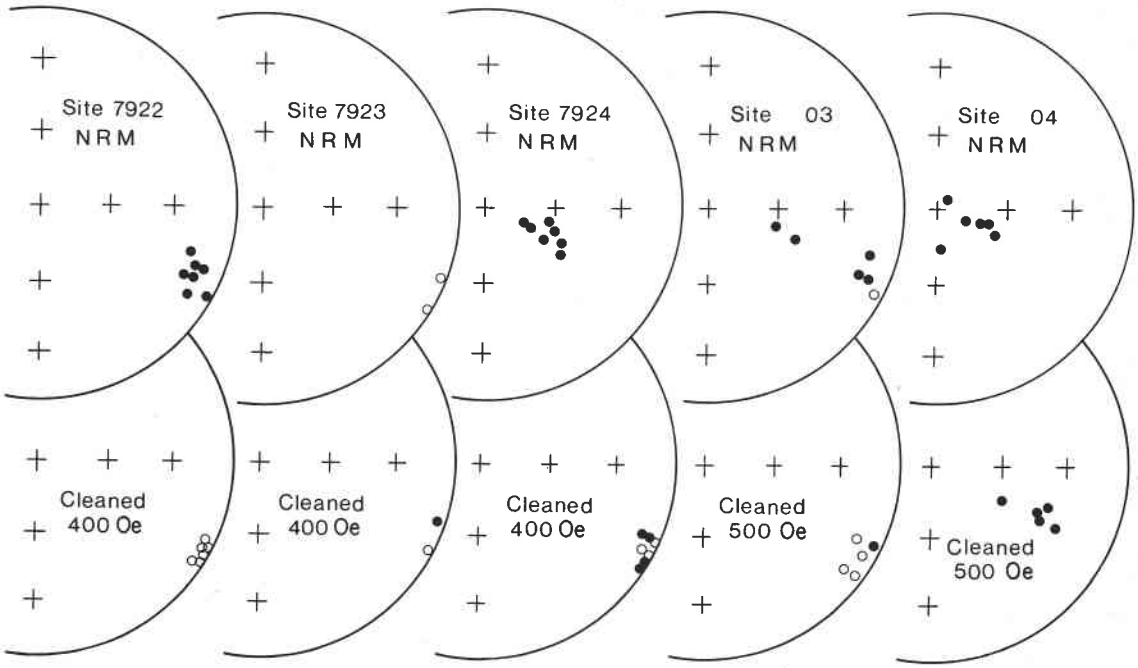


FIG. 5. AF paleomagnetic results from intersection II (Fig. 1).

Intersection II

The relationship at this intersection (Figs. 1, 4) is very similar to that at intersection I. At intersection II, however, although regionally the dykes trend at about 35° to one another, the small dyke cuts very obliquely through the older dyke. As a result of this, the younger dyke is only about 5 m thick within the older dyke. The paleomagnetic results (Fig. 5) are almost identical to those obtained for intersection I. An anomaly was encountered in processing samples of site 7924. Samples 01–08 were drilled in the narrow cross-cutting dyke and samples 09 and 10 were 15 m distant in the large dyke. Samples 09 and 10 were thought to be well beyond the point at which they are likely to have been remagnetized by the younger dyke, yet they moved out to the flat Franklin direction in response to AF demagnetization (points for 09 and 10 are not shown in Fig. 5). However, the response of these two samples to thermal demagnetization differed from that of samples taken in the thin cross-cutting dykes. Test samples from the little dyke essentially duplicated AF results, whereas samples 09 and 10 reacted more complexly, moving out part way then at higher temperatures returning to a steep (Borden dyke?) direction (Fig. 17). It is possible, because outcrop is not continuous, that a branch of the younger dyke or a continuation of the northwest trending dyke represented by site 7922 (Fig. 4) occurs near the collecting points of these cores and escaped observations. Chilled material from the narrow cross-cutting dyke (site 7924) yielded a whole-rock K–Ar age of 675 ± 46 Ma, K = 0.65, which is concordant with the age of Franklin material from intersection I.

Paleomagnetism of the Borden dykes

The two dyke intersections described above indicate that north Baffin is intersected by at least two major dyke swarms whose magnetization directions differ by about 60°. The younger of these, as indicated by cross-cutting relationships, has all of the characteristics ascribed to Franklin intrusions. The older swarm will be referred to as Borden dykes. These include the two thick northwest trending dykes with steep southeast magnetization directions at the previously described intersections. Paleomagnetic data from the two intersections indicate that the refraction of the younger (Franklin) dykes as they cross the zone of intense faulting (compare site 7917 with site 7920 and sites 7922 and 7923 with site 7924) was a primary intrusive feature. This is also indicated by the thinning of the cross-cutting dyke at intersection II, as it cuts obliquely across the older dyke.

Figure I indicates the paleomagnetic sites displaying the Borden direction, and results from these sites are summarized in Table 1. There is evidence of at least one Earth's field reversal during intrusion of the Borden dykes.

Fresh, chilled, marginal material from the Borden dykes represented by site 7908 yielded a whole-rock K–Ar age of 819 ± 107 Ma, K = 0.25. This is likely to

TABLE 1. Paleomagnetic data for the diabase dykes of Borden Peninsula

	NRM						Cleaned					
Site	*N*	*n*	*D*°	*I*°	K	α_{95}°	AF	*D*°	*I*°	P	K	α_{95}°
Borden dykes												
01	6	6	111	79	171	5	450	108	49	R	180	5
02	7	7	96	79	173	5	550	113	46	R	60	8
04	6	6	114	74	55	9	500	112	41	R	84	7
49	7	7	35	82	18	15	300	114	61	R	16	16
55	8	8	110	69	37	9	400	122	44	R	102	6
68	8	7	151	68	2	54	450	115	45	R	156	5
7904	8	8	103	83	233	4	300	122	62	R	72	7
7906	8	6	119	78	131	6	400	120	44	R	89	7
7908	8	7	92	85	137	5	400	104	65	R	132	5
7910	9	8	81	84	95	6	400	112	71	R	118	5
7912	8	7	13	83	417	3	400	90	66	R	106	6
7921	6	6	117	79	185	5	600	129	70	R	256	4
63	7	7	290	34	3	41	350	288	−44	N	43	9
66	7	6	318	73	102	7	500	311	−55	N	37	11
Mean normal direction (cleaned) of 14 sites above								294.7	−54.8		44	6
Mean pole position from mean direction							153.3°E	26.7°S	δm = 8.6°		δp = 6.1°	
Mean pole position from 14 site poles							152.8°E	28.1°S	K = 35.1		α_{95} = 6.8°	
Franklin dykes												
50	6	6	316	78	31	12	450	314	4	N	25	14
56	6	6	327	54	21	15	350	305	−15	N	34	12
59	6	5	306	55	30	14	400	303	0	N	91	8
61	8	8	300	70	19	13	500	311	13	N	47	8
03	6	5	110	30	13	22	500	118	−6	R	107	7
46	6	6	126	32	11	21	200	118	1	R	81	8
48	7	7	124	64	9	22	500	134	9	R	28	12
67	7	5	114	68	203	5	300	123	9	R	39	12
7917	8	8	131	42	86	6	400	130	0	R	120	5
7920	4	3	144	81	52	17	500	133	1	R	192	9
7922	8	7	113	14	126	5	400	119	−4	R	429	3
7923	2	2	115	−02	—	—	500	112	1	R	—	—
7924	8	7	114	62	102	6	400	115	1	R	206	4
Mean normal direction (cleaned) of 13 sites above								304.2	−0.9		63	5
Mean pole position from mean direction							153.4°E	09.2°N	δm = 5.3°		δp = 2.6°	
Mean pole position from 13 site poles							153.3°E	09.2°N	K = 118		α_{95} = 3.8°	
Tremblay Sound dyke												
45	8	7	311	60	477	3	500	304	32	N	466	3
7901	6	6	299	67	350	4	500	293	28	N	74	8
7902	7	6	308	66	298	4	400	302	37	N	108	7
7903	8	6	309	66	64	8	600	299	36	N	47	10
Mean normal direction (cleaned) of 4 sites above								299.3	33.5		207	6
Mean pole position from mean direction							165.8°E	26.1°N	δm = 7.3°		δp = 4.2°	
Mean pole position from 4 site poles							165.8°E	26.2°N	K = 232		α_{95} = 6°	
Steep overprint												
47	7	6	7	85	316	4	300	359	86	N	69	8
7915	6	6	44	80	110	6	300	136	88	N	38	11
7918	8	8	350	85	325	3	300	358	82	N	367	3
51	6	5	172	84	10	30	400	354	−78	R	16	6
Mean normal direction (cleaned) of 4 sites above								127.9	89.2		88	10
Mean pole position from mean direction							078.3°W	71.6°N	δm = 19.7°		δp = 19.7°	
Mean pole position from 4 site poles							078.4°W	71.8°N	K = 24		α_{95} = 19°	

TABLE 1. (*Concluded*)

Site	NRM						Cleaned					
	N	*n*	*D*°	*I*°	K	α_{95}°	AF	*D*°	*I*°	P	K	α_{95}°
Unclassified sites												
11	6	6	320	74	121	6	500	313	57	N	114	6
52	7	7	49	65	9	22	400	325	−10	N	14	17
60	8	8	255	84	25	11			Random			
62	7	7	339	82	58	8			Random			
64	7	7	80	76	53	8	450	197	−55	R	7	24
65	6	6	282	81	183	5			Random			

NOTES: Sites prefixed 79 were sampled in 1979, the remaining sites were sampled in 1977. *N* = number of cores collected at a site; *n* = number after eliminating those cores whose directions differed by more than three standard deviations from the site mean direction; *D* = declination; *I* = inclination; K = Fisher's estimate of precision; α_{95} = semi-axis of the circle of confidence at P = 0.95; AF = peak alternating field in oersteds; P = polarity, arbitrarily designated as follows: N = normal (north-seeking direction yields pole in Pacific), R = reversed; δm, δp = semi-axes of ellipse of confidence at $P = 0.95$.

be a minimum age for this intrusion; a figure of 950 Ma is suggested as a reference age for these dykes until a more precise figure is obtained. This figure was chosen because whole-rock K–Ar ages of chilled diabase are usually 10–15% less then the ages of intrusion. The dyke from which the 819 Ma age was obtained is defined as the "type" Borden dyke (Fig. 1) and it is recommended that in future work the definition and naming of dyke swarms include the designation of such a "type" dyke.

Paleomagnetism of the Franklin dykes of Baffin Island

Stable Franklin magnetization was isolated in 13 sites on dykes of the Borden Peninsula. Figure 1 locates sites displaying Franklin directions, and results from these sites are summarized in Table 1. Nine sites are reversely magnetized, and four sites normally.

Several sites exhibit evidence of both a normal and a reversed Franklin component. The thermally demagnetized specimens from sites 7917 and 7922 moved to and retained the Franklin AF direction, but at temperatures of 350–500°C exhibited major increases in total moment with no direction change (Fig. 6). No such increase in intensity was observed with AF cleaning; however, this may be a function of the nature of the two overlapping coercivity spectra, which can differ radically from the blocking temperature spectra. Intensity increase during thermal demagnetization probably indicates the presence of a reversed component. A thermally step-cleaned, chilled Franklin specimen from site 7920 shifted towards the AF reversed Franklin direction but at high temperatures moved towards a normal Franklin direction (Fig. 7). Specimens from this site also exhibited intensity increases during step AF cleaning at fields of 300–400 Oe (23.9–31.8 kA/m).

The suggested age of the Franklin intrusions based on the two new whole-rock K–Ar analyses of chilled material (649 ± 42, 675 ± 46 Ma), and on the earlier reported K–Ar ages is 750 Ma. This is about 100 Ma less than the age suggested by Berger and York (1980) in calibrating the younger part of the Grenville Loop. Experience over a period of 20 years with whole-rock K–Ar ages of chilled basic material that has subsequently been dated by the Rb/Sr method makes their greater age seem unlikely.

The magnetization site directions for the Borden and Franklin dykes of north Baffin are shown in Fig. 8.

Other dyke sites

A number of sites cannot be classified paleomagnetically as either Franklin or Borden.

Group 1—the Tremblay Sound dyke

This group consists of sites 45 and 7901, 7902, 7903 (Fig. 1; Table 1). These have well grouped, stable, northwest down directions (Fig. 9) and may represent a single, en échelon dyke with a trend usual for the Borden dykes. There are a number of possible explanations for the magnetization direction. The pole position is north of the Franklin pole for Borden Peninsula. It could therefore reflect post-Franklin remagnetization during a later thermal event. This is unlikely because there is no evidence of such an event in neighbouring dykes or sediments, nor is there evidence of two components in any of the samples, except for a viscous component probably related to the recent Earth's field. A second possibility, the presence of both normal and reversed components because of a field reversal during cooling of the dyke, also seems unlikely because of the homogeneity of magnetization for 40 km along a dyke whose width varies by a factor of two. The best explanation seems to be the intrusion of a dyke along the Borden fault zone after the main Franklin intrusive episode.

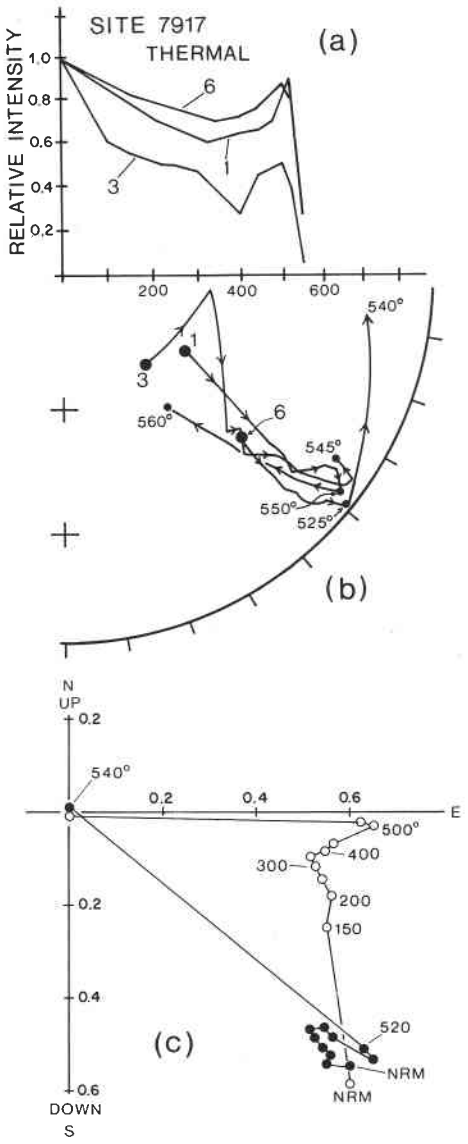


FIG. 6. Thermal demagnetization of three samples from site 7917 suggesting the presence of both reversed and normal components. (*a*) Relative intensity and (*b*) direction changes during heatings to 550°C. (*c*) Zijderveld diagram for demagnetization of sample 6. In this and subsequent Zijderveld diagrams, open circles represent orthogonal projections in the vertical (east–west) plane, solid circles in the horizontal plane.

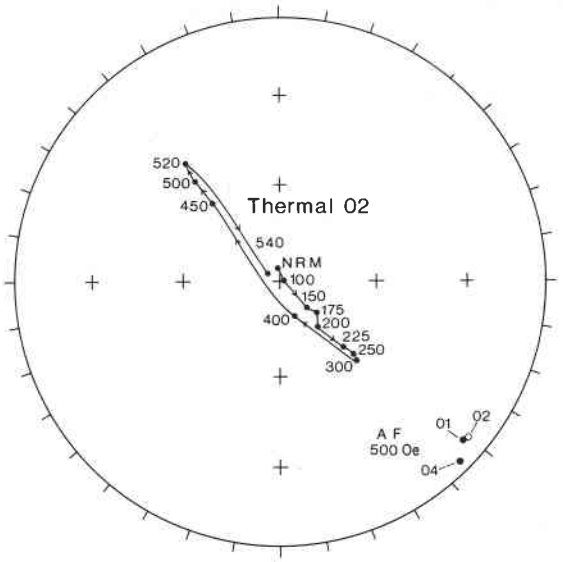


FIG. 7. Detailed thermal cleaning of specimen 02 from site 7920 suggesting the presence of normal and reversed components. AF cleaning does not provide similar evidence.

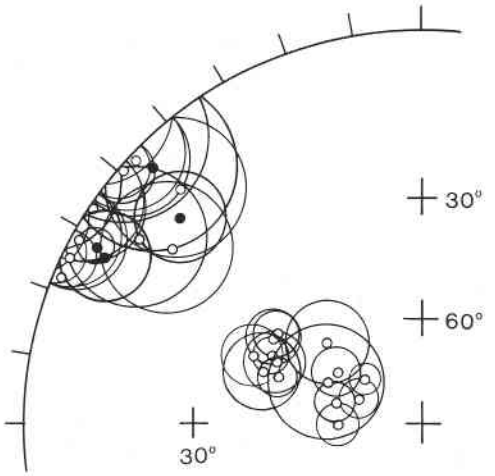


FIG. 8. Mean site directions with circles of confidence, of Borden and Franklin dykes from north Baffin. Where necessary south-seeking directions have been used so that all projection appear in the northwest quadrant.

Site 11 (Figs. 1, 10) on a dyke with an anomalous northeast strike yielded a pole similar to that of the Tremblay Sound dyke. This may be a second post-Franklin intrusion.

Group 2—very steeply overprinted

This group consists of four sites, 47, 51, 7915, and 7918 (Fig. 1), two of which, 47 and 7915, are from the same dyke. The four sites are located on dykes that are particularly thick (at least 100 m), very coarse grained, and rather weathered. At site 7918 (Fig. 2) the dyke consists essentially of very coarse-grained granophyric material. The dyke contacts are not exposed, but the granophyric zone is bounded by 20 m wide zones of crumbly, khaki–grey material. Site 7921, located on the same dyke >60 m from the intersection with the narrow dyke (Fig. 2), represents more typical diabase and has a direction more typical of the Borden dykes. Sites 47 and 7915 retained a steeply down direction in response to AF cleaning to a field of 350 Oe (27.8 kA/m), above which

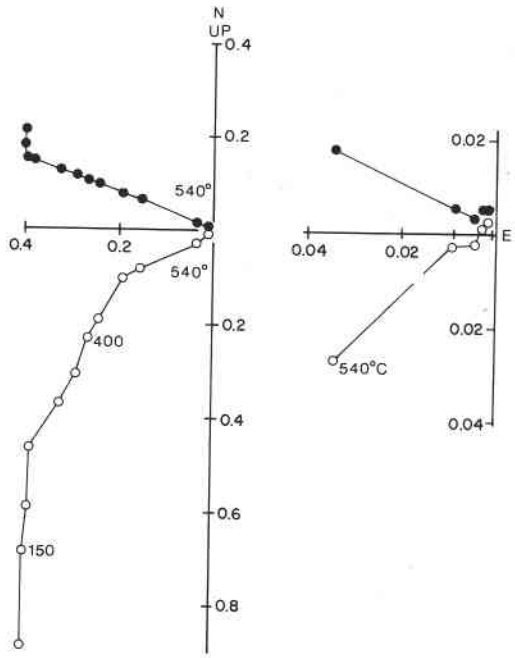


FIG. 9. Zijderveld diagram of thermal demagnetization of a sample from site 7901 (Tremblay Sound dyke). The right-hand diagram is an enlargement of the higher temperature (540–580°C) part of the plot.

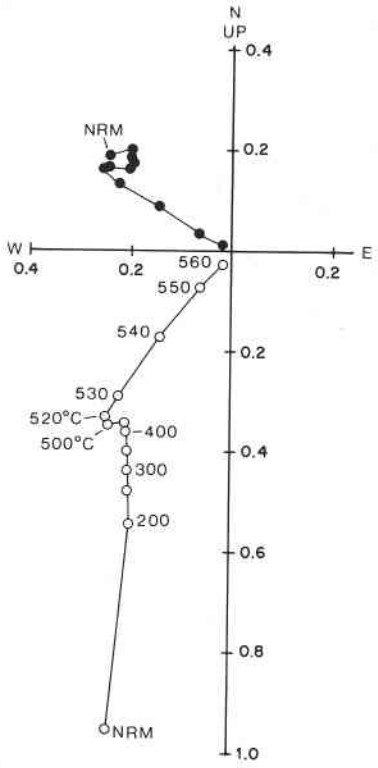


FIG. 10. Zijderveld diagram of thermal demagnetization of specimen from site 11, which has an AF cleaned direction similar to that of the Tremblay Sound dyke.

they were unstable. During thermal step cleaning they retained a very steeply down direction to 570 and 540°C, respectively. The data on these dykes suggest the presence of a component related to their coarse-grained, easily weathered character and it is suggested that this is a chemical remanent magnetization (CRM) component with a high unblocking temperature. Evidence for such a component has been identified in several other Borden Peninsula dykes, particularly in their responses to thermal cleaning, and a situation similar to the one described above was also encountered in a study of the Aston dykes of Somerset Island (Jones and Fahrig 1978). In this case, a very wide granophyric dyke, in a group of Franklin dykes of identical strike, exhibited a very stable steep direction. It was suggested that the dyke might be a Tertiary intrusion but the close association with Franklin dykes, and our data from Borden Peninsula, make this unlikely. Some examples of evidence, particularly in response to thermal cleaning, for a CRM component are now given for other sites on Borden Peninsula.

(1) The magnetization directions of two specimens from site 01 shifted during thermal cleaning towards the AF Borden end-point direction, but about 450°C again developed a steeper inclination. The vector removed up to 400°C was very steep.

(2) The direction of one specimen from Franklin site 03, under thermal cleaning, shifted part way to the flat Franklin direction, then above 300°C steepened (Fig. 11). A second specimen shifted to the Franklin direction and remained there (Fig. 12).

(3) Specimens from site 47 retained a very steeply down direction under AF demagnetization to 350 Oe (27.8 kA/m), above which they were unstable. Their directions remained very steeply down under thermal demagnetization to 570°C.

(4) The magnetic inclination of cores from site 60 changed from steeply down to dispersed up, under AF cleaning. Two of five thermally cleaned specimens yielded a steeply up direction (500 and 520°C) but the remaining three specimens retained a steeply down direction.

(5) Site 66 yielded a stable up (normal Borden) AF direction. Three thermally demagnetized specimens shifted towards this Borden direction but at high temperature returned to a very steeply down direction (Figs. 13, 14). The vector removed at 350–450°C was normal Borden but at higher temperatures was steeply down. This indicates the presence of a steeply down component with a high unblocking temperature.

(6) Site 62 did not yield a stable direction in response

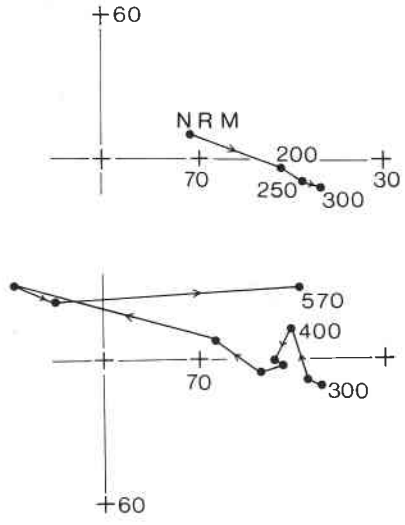


FIG. 11. Stereographic projection of magnetization directions during step thermal cleaning of sample 4 from Franklin site 03. Note a shift towards a Franklin direction at low temperature with a return to steep (CRM?) direction above 300°C. Note that the two parts of the diagram overlap but are shown separately for clarity.

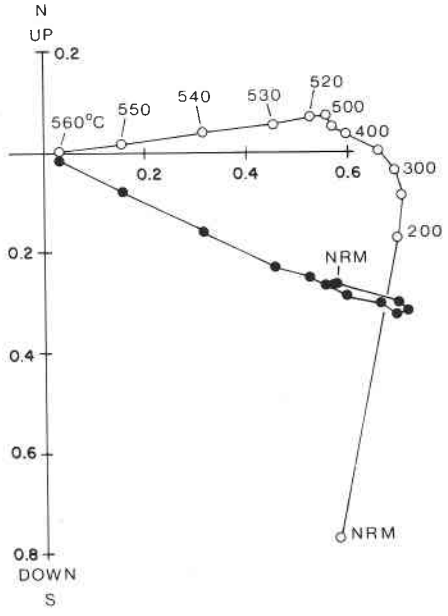


FIG. 12. Zijderveld diagram of thermal demagnetization of sample 2 from Franklin site 03. Compare with directional changes of sample 4 (Fig. 11) at higher temperatures. Note that this sample shows no evidence of a steep (CRM?) direction above 300°C.

to AF cleaning, and remained steeply down at temperatures up to 450°C.

(7) Five samples from site 67 yielded a stable reversed Franklin direction but two maintained a steep easterly direction, suggesting a steep overprint. Thermal cleaning resulted in a shift towards the Franklin direction but a return to steeper directions at high temperatures (Fig. 15).

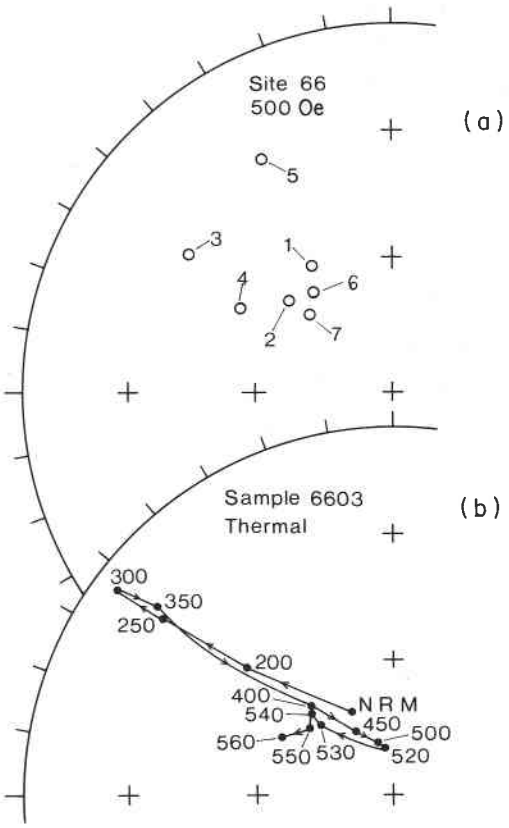


FIG. 13. (*a*) Magnetization directions of samples from site 66 after AF cleaning at 500 Oe. (*b*) Magnetization directions of sample 3 during thermal cleaning. Note that sample direction moves towards (without reaching) the end-point AF direction but returns to a steeply down direction at high temperature.

(8) A specimen from Borden site 7912 retained a distinctly steeper direction in response to thermal cleaning than in response to AF cleaning.

(9) Three specimens of the anomalous site 7918 retained a steep magnetic direction to 570°C, whereas a fourth specimen shifted to a Borden direction in the range 520–550°C.

(10) The direction of the baked Borden dyke site 7919 (Fig. 3) shifted to the Franklin direction at moderate temperatures, but at high temperatures two specimens returned to a very steep direction throughout the range 560–600°C (Fig. 16).

(11) Chilled Franklin site 7920 moved towards, but not close to, a reversed Franklin direction at lower temperatures, shifted towards a normal Franklin direction at moderate temperatures, then to a very steep direction at highest temperatures.

(12) The magnetization of samples 01–08 of site 7924 (Figs. 4, 5), taken from the small cross-cutting

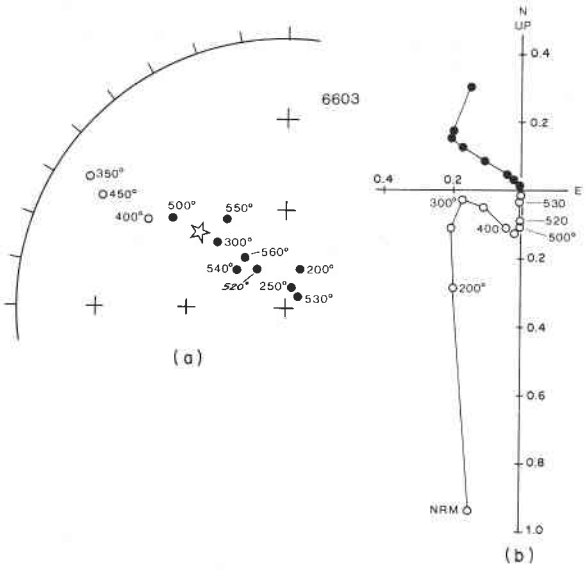


FIG. 14. (*a*) Vector removed during cleaning of sample 3, site 66 (see also Fig. 13). The open star is the site-mean AF end-point direction. (*b*) Zijderveld diagram of thermal cleaning of sample 3.

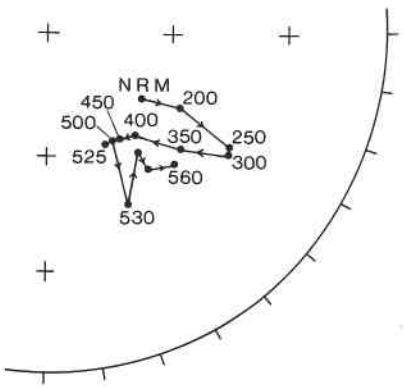


FIG. 15. Directional changes during thermal cleaning of sample 2 from site 67. Five of seven samples from this site moved to a stable, shallow southeast (reversed Franklin) direction as a result of AF cleaning. Thermal cleaning of this sample suggests the presence of both a Franklin and a steeper component.

dyke, shifted to a Franklin direction with thermal cleaning, but sample 09 from the older dyke (but well away from the narrow dyke) shifted at low temperatures towards the Franklin direction and at high temperatures back to a steep direction (Fig. 17).

Interpretation

The very steeply down component documented above is interpreted as an overprint, and as it has affected Franklin as well as Borden dykes it has an age less than 750 Ma. This overprint has a direction that could fit in

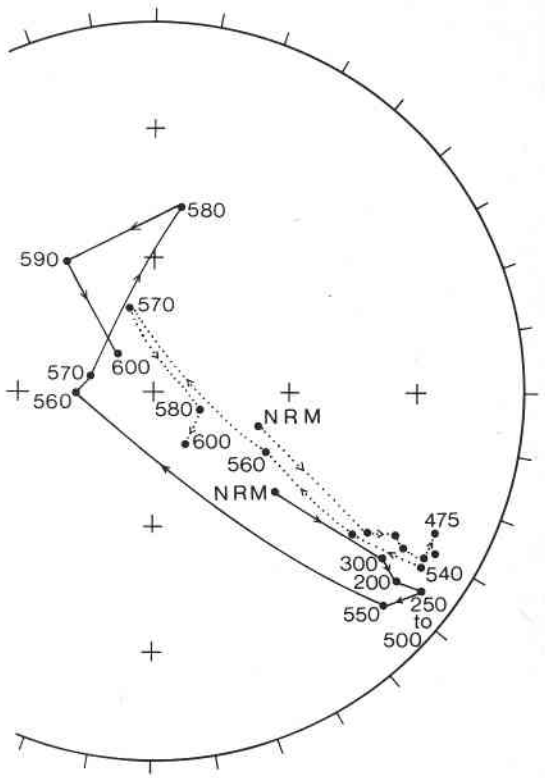


FIG. 16. Thermal cleaning of two baked specimens. They shift to a reversed Franklin direction at low temperatures but return to a very steep direction at high temperatures, possibly indicating a steep CRM overprint.

one of two age brackets—either late Hadrynian to early Paleozoic or late Mesozoic to Cenozoic. It was suggested in the first study of Franklin dykes that such an overprint could be related to mild heating (and oxidation?) related to the opening of Baffin Bay (Fahrig *et al.* 1971). It is not possible at present to choose between this and a possibly earlier age for the overprinting.

Group 3—steeply up to moderately south up direction

The specimens of site 64 exhibit fairly stable end points with southerly up directions in response to AF demagnetization. This component does not appear related to directions of either Franklin or Borden dykes. Detailed AF results from site 64 are shown in Fig. 18. Samples 01–05 and 07 exhibit good end points, although within-site directions are dispersed (D = 197°, I = 55°, K = 7) at a cleaning field of 450 Oe (35.8 kA/m). Two samples (04, 06) exhibit a steeply northwest up direction at 500 Oe (39.8 kA/m), which could indicate the presence of a normal Borden direction, but no high field end point was established because all samples show large, random directional changes above 500 Oe (39.8 kA/m).

The orthogonal projections of the demagnetization paths of samples 01–05 and 07 are linear over a

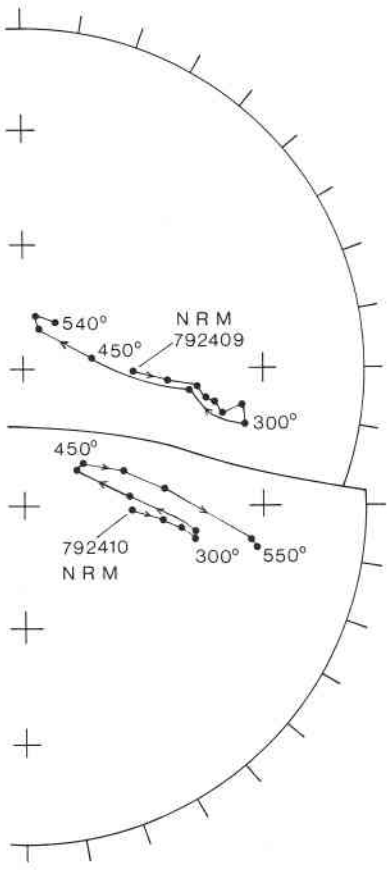


FIG. 17. Stereographic plot showing directional changes as a result of step thermal demagnetization of two specimens from site 7924 indicating a shift towards a Franklin or a Borden direction at low temperatures and a return to a steep, possibly CRM overprint direction at intermediate or high temperatures.

considerable intermediate AF range and those of 02, 03, 05, and 07 are linear up to 500 Oe (39.8 kA/m). This linearity does not necessarily indicate the isolation of a magnetic component. If two hard components are present they may come off in a proportion such that the demagnetization trajectory is linear and neither component is isolated.

Thermal demagnetization gave results very roughly similar to AF results. The one specimen from site 52 that was thermally treated displayed a rather unstable but steeply up direction above 500°C. Two of five thermally demagnetized samples from site 60 gave a steeply up direction at 500 and 520°C and the remaining three stayed steeply down. Two thermally demagnetized samples from site 64 yielded a south up direction in the range 500–530°C.

Perhaps the best explanation of these up directions is that they indicate a reversed CRM component of the type previously discussed. It would be quite possible for this component and the normal component to be present together as a result of a field reversal during the imposition of a CRM. It would also be quite possible for both a normal and reversed component to be present in the same core and overlapping AF coercivity and thermal unblocking spectra so that they could not be entirely separated.

Discussion of the Borden pole

Data have been provided suggesting that some diabase on Borden Peninsula has acquired a steeply inclined later component, so it is reasonable to ask whether all the "Borden" dykes might simply be Franklin dykes containing a mixture of Franklin and Tertiary or recent Earth's field components as suggested in earlier papers (Fahrig *et al.* 1971; Fahrig and Schwarz 1973). There are a number of good reasons for discounting such a possibility. First, we have excellent cross-cutting relationships at the two previously described intersections and an excellent baked contact test at intersection 1. Chilled material from the two younger dykes yielded K–Ar ages of 649–675 Ma, whereas chilled material from a dyke with a steeply down (Borden) direction yielded an age of 819 Ma. This last figure suggests an age greater than 900 Ma, too old for this dyke to be Franklin. The evidence of two normally polarized (steeply northwest up) sites (63 and 66) among the 12 sites with Borden directions (Table 1; Fig. 19) is difficult to explain in terms of a mixed Franklin and young component. Sites 60 and 62, though dispersed, also showed evidence of a steeply up component. These four sites are only readily explained in terms of reversed and normal Borden dyke directions.

The Borden pole is particularly important in the study of Helikian – early Hadrynian polar wandering for North America because North American poles of this age from rocks outside the Grenville Province are very rare. For example, Berger and York (1980) in a thoughtful review of North American apparent polar wandering for the period 1500–800 Ma have defined the southern extremity of the Grenville Loop on the basis of poles from only the Grenville Province (Fig. 20). Unfortunately, the position of the Grenville Province relative to bordering older provinces during this period has not been firmly established, nor has the age of many of these Grenville poles.

These uncertainties are less pronounced regarding the early Hadrynian Borden dyke pole. It has been demonstrated that poles from the lower and upper Borden sedimentary–volcanic sequence are 1220–1200 Ma old and that north Baffin has not shifted significantly relative to the mainland Canadian Shield since this time (Fahrig *et al.* 1981). Because the Borden and Franklin dykes cut the Borden sequence they cannot have moved relative to older mainland rocks and their poles should provide

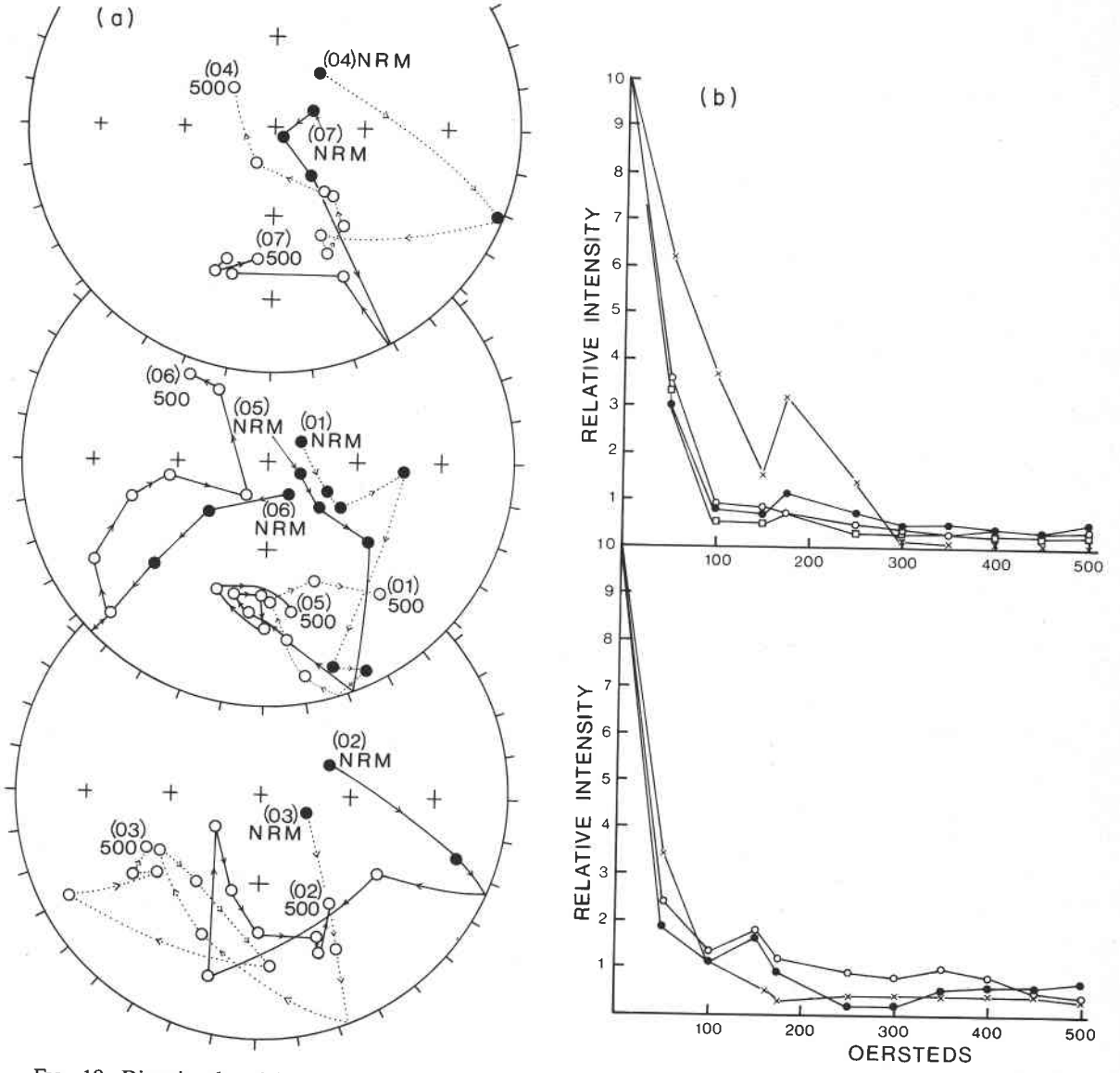


FIG. 18. Directional and intensity changes in the seven samples of site 64 in response to AF cleaning. Note the unusually large component of low coercivity.

reliable information on the apparent polar wandering path for the Canadian Shield during the late Helikian – Hadrynian.

We now have available from north Baffin four sets of poles, the lower Borden Basin pole of 1220 Ma (data from the Adams Sound and Nauyat volcanics), the upper Borden Basin pole (Strathcona Sound) about 20 Ma years younger, the Borden dykes pole at about 950 Ma, and the Franklin dykes pole at about 750 Ma. As the metamorphic grade of these rocks is sub-greenschist, it is extremely unlikely that any of them has been overprinted by younger TRM.

In Fig. 20 is shown part of the Berger and York (1980) apparent polar wandering curve for the period 1500–800 Ma, plotted here in the Pacific rather than in the antipodal Atlantic position as in Berger and York. Superimposed are the four poles from Borden Peninsula, Baffin Island. All are in acceptable agreement, in both position and age, with Berger and York's proposed path. The Borden dyke pole, age circa 950 Ma, is younger than the upper Borden Basin sedimentary (Strathcona Sound) pole and older than the Franklin pole. It is therefore bracketed in such a way that it must lie on the Grenville Loop even if the age accepted here is slightly low. The Borden pole therefore provides an anchor on the Grenville Loop similar to those provided

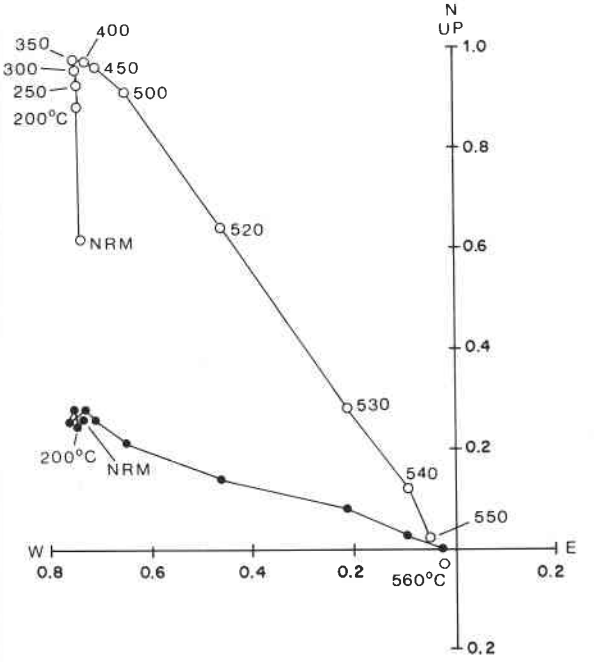


FIG. 19. Zijderveld plot of thermally demagnetized sample from normally magnetized Borden dyke site 63.

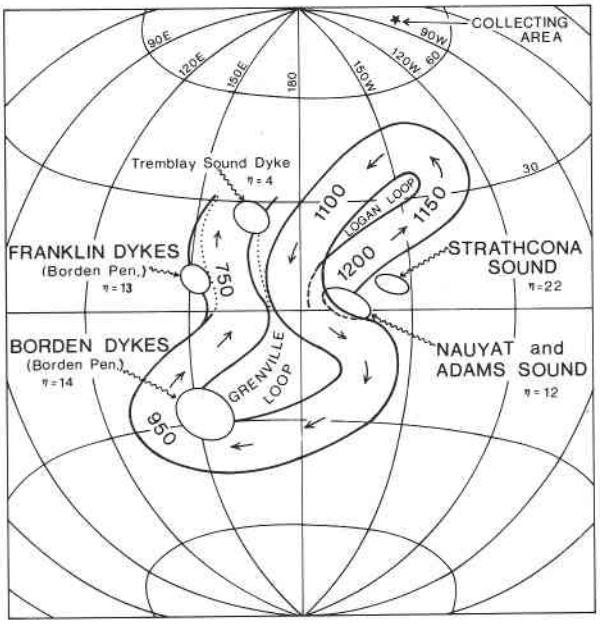


FIG. 20. North American polar wandering curves from 1220 to ~700 Ma, modified slightly from Berger and York (1980).

for the Logan Loop (Robertson and Fahrig 1971). In Fig. 20 the apparent polar wandering curve in the vicinity of the Franklin poles has been slightly modified to better accommodate both the new Franklin data and the older published data.

Tectonic implications of Borden Peninsula dykes

The pattern of dykes on Borden Peninsula (Fig. 1) suggests the presence of two major tensional regimes of significantly different age. Because the northwest dykes are parallel to the pre-existing Helikian fault zone, one would speculate that this is the older of the two, and this is supported by the tendency of the more northerly striking dykes to be deflected as they cross the major zone of northwest dykes. This relative age is also supported by the previously described intersections. This simple picture of two dyke swarms with different ages and directions indicating two periods and directions of tension is not, however, supported by paleomagnetic work. Figure 1 shows Borden (950 Ma) and Franklin (750 Ma) dykes identified on the basis of paleomagnetic results. It is likely that most of the wide northwest trending dykes of Borden Peninsula are Borden dykes, and most of the north-northwest or north trending dykes are Franklin, but both trends are represented by Borden and Franklin dykes. This suggests that the dykes reflect two magmatic pulses separated by about 200 Ma, with both pulses taking place under east-northeast tensional conditions. The northwest dyke system probably resulted from dilation along pre-existing Helikian (1200 Ma) fault zones during the two periods of east-northeast tension. It further suggests that the present surface of erosion was at a shallow depth during both intrusive episodes.

These two episodes of east-northeast tension may be related to the early development of the Boothia Uplift. The postulated east-northeast tensional direction is roughly at right angles to the axis of Boothia Arch, and Kerr (1977) has referred to two periods of arching that probably occurred during the Late Proterozoic. These periods of arching may coincide with the intrusion of Borden dykes (950 Ma) and Franklin dykes (750 Ma).

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